

Evaluate the following geometric sums or state that it diverges.

1. $\sum_{k=0}^4 3^k$

2. $\sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^{-k}$

3. $\sum_{k=0}^{\infty} \frac{\pi}{4^k}$

4. $\sum_{k=0}^{\infty} \frac{3 \cdot 4^k}{7^k}$

5. $\sum_{k=2}^{\infty} 3(-0.75)^k$

6. $\sum_{k=1}^{\infty} e^{-2k}$

7. $\sum_{k=1}^{\infty} \frac{3^{k-1}}{4^{k+1}}$

For the following telescoping series, find a formula for the n^{th} term of the sequence of partial sums $\{S_n\}$. Then evaluate the $\lim_{n \rightarrow \infty} S_n$ to obtain the value of the series or state that it diverges.

8. $\sum_{k=1}^{\infty} \left(\frac{1}{k+1} - \frac{1}{k+2} \right)$

9. $\sum_{k=1}^{\infty} \left(\frac{1}{k+2} - \frac{1}{k+3} \right)$